Package insert

Montas-10 (Montelukast Film Coated Tablets 10mg)

1. Name of the medicinal product
Montas-10 (Montelukast Film Coated Tablets 10mg)

2. Qualitative and quantitative composition
Each film-coated tablet contains Montelukast sodium Ph.Eur. equivalent to Montelukast 10 mg
For a full list of excipients, see section 6.1.

3. Pharmaceutical form
Montas 10 is beige coloured, rounded square, biconvex, film coated tablet debossed "M10"on one side and plain on other side, which is packed in aluminium-aluminium blister pack containing printed aluminium blister foil on one side and plain aluminium blister foil on other side.

4. Clinical particulars
4.1 Therapeutic indications
For the prophylaxis and chronic treatment of asthma in adults and adolescents 15 years of age and older. Montas 10 is indicated in adults and adolescents 15 years of age and older for the relief of daytime and night-time symptoms of seasonal allergic rhinitis.

4.2 Posology and method of administration
Posology:
Montas 10 should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.
Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Adults 15 Years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis
The dosage for adults 15 years of age and older is one 10-mg tablet daily.

General Recommendations
The therapeutic effect of Montas 10 on parameters of asthma control occurs within one day. Montas 10 tablets can be taken with or without food. Patients should be advised to continue taking Montas 10 while their asthma is controlled, as well as during periods of worsening asthma.

No dosage adjustment is necessary for pediatric patients, for the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent.

Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with Montas 10 in Relation to Other Treatments for Asthma
Montas 10 can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:
Bronchodilator Treatments: Montas 10 can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with Montas 10 provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. Montas 10 should not be abruptly substituted for inhaled corticosteroids.

Method of administration:
Oral use.

4.3 Contraindications
Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use
Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β-agonist should be used. Patients should seek their doctor's advice as soon as possible if they need more inhalations of short-acting β-agonists than usual.
Montelukast should not be substituted abruptly for inhaled or oral corticosteroids.

There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly.
In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases have been sometimes associated with the reduction or withdrawal of oral corticosteroid therapy. Although a causal relationship with leukotriene receptor antagonism has not been established, physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.

4.5 Interaction with other medicinal products and other forms of interaction
Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma.

The recommended clinical dose of Montelukast do not have clinically important effects on the pharmacokinetics of the following drugs: Theophylline, Prednisone, Prednisolone, oral contraceptives (Ethinyl estradiol/ Norethindrone 35/1), Terfenadine, Digoxin and Warfarin.
Since Montelukast is metabolized by CYP 3A4, caution should be coadministered with inducers of CYP 3A4, such as Phenytoin, Phenobarbital and Rifampicin.

Montelukast is not anticipated to markedly alter the metabolism of drugs metabolized by this enzyme (e.g. Paclitaxel, Rosiglitazone, and Repaglinide).

4.6 Fertility, pregnancy and lactation
Pregnancy
Animal studies do not indicate harmful effects with respect to effects on pregnancy or embryonal/foetal development.
Limited data from available pregnancy databases do not suggest a causal relationship between Montelukast 10 mg film-coated tablets and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience.

Montelukast 10 mg film-coated tablets may be used during pregnancy only if it is considered to be clearly essential.

Breastfeeding
It is unknown whether montelukast is excreted in human milk. Studies in rats have shown that montelukast is excreted in milk.
Montelukast 10 mg film-coated tablets may be used in breast-feeding only if it is considered to be clearly essential

4.7 Effects on ability to drive and use machines
Montas 10 has no or negligible influence on the ability to drive and use machines. However, individuals have reported drowsiness or dizziness.

4.8 Undesirable effects
Montelukast has been evaluated in clinical studies as follows:
- 10 mg film-coated tablets in approximately 4000 adult and adolescent asthmatic patients 15 years of age and older.
- 10 mg film-coated tablets in approximately 400 adult and adolescent asthmatic patients with seasonal allergic rhinitis 15 years of age and older.
- 5 mg chewable tablets in approximately 1750 paediatric asthmatic patients 6 to 14 years of age.

The following drug-related adverse reactions in clinical studies were reported commonly (1/100 to <1/10) in asthmatic patients treated with montelukast and at a greater incidence than in patients treated with placebo:

Body system Class	Adult and Adolescent Patients 15 years and older (two 12-week studies; n=795)	Paediatric Patients 6 to 14 years old (one 8-week study; n=201) (two 56-week studies; n=615)
Nervous system disorders	headache	headache
Gastrointestinal disorders	abdominal pain	

With prolonged treatment in clinical trials with a limited number of patients for up to 2 years for adults, and up to 12 months for paediatric patients 6 to 14 years of age, the safety profile did not change.

Tabulated list of Adverse Reactions
Adverse reactions reported in post-marketing use are listed, by System Organ Class and specific Adverse Experience Term, in the table below. Frequency Categories were estimated based on relevant clinical trials.

System Organ Class	Adverse Experience Term	Frequency Category*
Infections and infestations	upper respiratory infection†	Very Common
Blood and lymphatic system disorders	increased bleeding tendency	Rare
	thrombocytopenia	Very Rare
Immune system disorder	hypersensitivity reactions including anaphylaxis	Uncommon
	hepatic eosinophilic infiltration	Very Rare
Psychiatric disorders	dream abnormalities including nightmares, insomnia, somnambulism, anxiety, agitation including aggressive behaviour or hostility, depression, psychomotor hyperactivity (including irritability, restlessness, tremor§)	Uncommon
	disturbance in attention, memory impairment , tic	Rare
	hallucinations, disorientation, suicidal thinking and behaviour (suicidality), Dysphemia	Very Rare
Nervous system disorder	dizziness, drowsiness paraesthesia/hypoesthesia, seizure	Uncommon
Cardiac disorders	palpitations	Rare
Respiratory, thoracic and mediastinal disorders	epistaxis	Uncommon
	Churg-Strauss Syndrome (CSS), pulmonary eosinophilia	Very Rare
Gastrointestinal disorders	diarrhoea‡, nausea‡, vomiting‡	Common
	dry mouth, dyspepsia	Uncommon
Hepatobiliary disorders	elevated levels of serum transaminases (ALT, AST)	Common
	hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury).	Very Rare
Skin and subcutaneous tissue disorders	rash‡	Common
	bruising, urticaria, pruritus	Uncommon
	angiooedema	Rare
	erythema nodosum, erythema multiforme	Very Rare
Musculoskeletal, connective tissue and bone disorders	arthralgia, myalgia including muscle cramps	Uncommon
Renal and urinary disorders	enuresis in children	Uncommon
General disorders and administration site conditions	pyrexia‡	Common
	asthenia/fatigue, malaise, oedema,	Uncommon

* Frequency Category: Defined for each Adverse Experience Term by the incidence reported in the clinical trials data base: Very Common (≥1/10), Common (≥1/100 to </10), Uncommon (≥1/1000 to </100), Rare (≥1/10,000 to </1000), Very Rare (</10,000).
† This adverse experience, reported as Very Common in the patients who received montelukast, was also reported as Very Common in the patients who received placebo in clinical trials.
‡ This adverse experience, reported as Common in the patients who received montelukast, was also reported as Common in the patients who received placebo in clinical trials.
§ Frequency Category: Rare

Postmarketing Experience
Blood and lymphatic system disorders: thrombocytopenia

Psychiatric disorders:
Obsessive-compulsive symptoms

4.9 Overdose
In chronic asthma studies, montelukast has been administered at doses up to 200 mg/day to adult patients for 22 weeks and in short term studies, up to 900 mg/day to patients for approximately one week without clinically important adverse experiences.

There have been reports of acute overdose in post-marketing experience and clinical studies with montelukast. These include reports in adults and children with a dose as high as 1000 mg (approximately 61 mg/kg in a 42 month old child). The clinical and laboratory findings observed were consistent with the safety profile in adults and paediatric patients.

Symptoms of overdose
There were no adverse experiences in the majority of overdose reports. The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting and psychomotor hyperactivity.

Management of overdose
No specific information is available on the treatment of overdose with montelukast. It is not known whether montelukast is dialysable by peritoneal- or haemo-dialysis.

5. Pharmacological properties
5.1 Pharmacodynamic properties
Pharmacotherapeutic group: Leukotriene receptor antagonist
ATC code: R03D C03

Mechanism of action
The cysteinyl leukotrienes (LTC_4, LTD_4, LTE_4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 ($CysLT_1$) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Pharmacodynamic effects
Montelukast is an orally active compound which binds with high affinity and selectivity to the $CysLT_1$ receptor. In clinical studies, montelukast inhibits bronchoconstriction due to inhaled LTD_4 at doses as low as 5 mg. Bronchodilation was observed within 2 hours of oral administration. The bronchodilation effect caused by a β agonist was additive to that caused by montelukast. Treatment with montelukast inhibited both early- and late phase bronchoconstriction due to antigen challenge. Montelukast, compared with placebo, decreased peripheral blood eosinophils in adult and paediatric patients. In a separate study, treatment with montelukast significantly decreased eosinophils in the airways (as measured in sputum) and in peripheral blood while improving clinical asthma control.

5.2 Pharmacokinetic properties
Absorption
Montelukast is rapidly absorbed following oral administration. For the 10 mg film-coated tablet, the mean peak plasma concentration (Cmax) is achieved 3 hours (Tmax) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and Cmax are not influenced by a standard meal. Safety and efficacy were demonstrated in clinical trials where the 10 mg film-coated tablet was administered without regard to the timing of food ingestion.

Distribution
Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 litres. Studies in rats with radiolabelled montelukast indicate minimal distribution across the blood-brain barrier. In addition, concentrations of radiolabelled material at 24 hours post-dose were minimal in all other tissues.

Biotransformation
Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children.
Cytochrome P450 2C8 is the major enzyme in the metabolism of montelukast. Additionally CYP 3A4 and 2C9 may have a minor contribution, although itraconazole, an inhibitor of CYP 3A4, was shown not to change pharmacokinetic variables of montelukast in healthy subjects that received 10 mg montelukast daily. Based on *in vitro* results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

Elimination
The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabelled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

Characteristics in patients
No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Studies in patients with renal impairment have not been undertaken. Because montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. With high doses of montelukast (20- and 60-fold the recommended adult dose), decrease in plasma theophylline concentration was observed. This effect was not seen at the recommended dose of 10 mg once daily.

6. Pharmaceutical particulars
6.1 List of excipients
Tablet core:
Microcelac 100 (Lactose monohydrate + Cellulose microcrystalline),
Low substituted hydroxypropylcellulose (LH-11),
Croscarmellose sodium,
Magnesium stearate

Film coat:
Hydroxypropylcellulose (LF),
Hypromellose 6CPS (A),
Titanium dioxide,
Macrogol 6000,
Iron oxide yellow,
Iron oxide red
Purified water

6.2 Incompatibilities
Not applicable.

6.3 Shelf life
2 years

6.4 Special precautions for storage
Do not store above 30°C. Store in the original package in order to protect from light and moisture.

6.5 Nature and contents of container
Montelukast 10 mg tablets are packed in Aluminium-Aluminium blister.
Pack size: Packs of 14 tablets in blister.

7. Instructions for Use
Montas 10 tablets can be taken with or without food.
Patients should be advised to continue taking Montas 10 while their asthma is controlled, as well as during periods of worsening asthma.

8. Product Registration Holder
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8. Manufacturer
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8. Date of revision of the text
April 2024

INP113
RMON01I2608

Front Side

Back Side

Text matter - Arial Narrow 9 pt
Brand Name - Arial Black 11 pt

RMON01I2608 - Montas 10-Malaysia-PIL
PIL Size: 170 x 550 mm (Front/Back side) (Text area 160 x 540 mm)
Colour: Pantone Black
Date: 18*07*23, 19*07*23, 26*10*23, 01*02*24, 13-03-2024